

Advanced (Carolina Reaper)

- Q1.** Factor the trinomial $x^2 + 5x + 6$.
(A) $(x + 2)(x + 3)$
(B) $(x + 1)(x + 4)$
(C) $x(x + 5) + 6$
(D) $(x + 1)(x + 2)$
(E) $x(x + 6) + 5$
- Q2.** Tom has been saving money for a new bike and has $x^2 + 4x + 4$ dollars. Factor the expression to determine the number of weeks Tom has been saving.
(A) $x^2 + 4x + 4 = (x + 2)^2$
(B) $x^2 + 4x + 4 = (x + 1)^2$
(C) $x^2 + 4x + 4 = x(x + 4)$
(D) $x^2 + 4x + 4 = x^2 + 4x$
(E) $x^2 + 4x + 4 = x + 4$
- Q3.** A bookshelf has $x^2 - 7x + 12$ books on it. Factor the expression to find the number of shelves.
(A) $x^2 - 7x + 12 = (x - 3)(x - 4)$
(B) $x^2 - 7x + 12 = (x - 2)(x - 5)$
(C) $x^2 - 7x + 12 = x(x - 7) + 12$
(D) $x^2 - 7x + 12 = (x - 1)(x - 6)$
(E) $x^2 - 7x + 12 = x(x - 7)$
- Q4.** Factor the trinomial $x^2 - 2x - 15$.
(A) $x^2 - 2x - 15 = (x - 5)(x + 3)$
(B) $x^2 - 2x - 15 = (x + 5)(x - 3)$
(C) $x^2 - 2x - 15 = (x + 1)(x - 15)$
(D) $x^2 - 2x - 15 = (x - 1)(x + 15)$
(E) $x^2 - 2x - 15 = x(x - 2) - 15$
- Q5.** Ashley is planning a road trip and wants to drive $x^2 + x - 6$ miles. Factor the expression to determine the number of hours she needs to drive.
(A) $x^2 + x - 6 = (x + 3)(x - 2)$
(B) $x^2 + x - 6 = (x + 1)(x - 6)$
(C) $x^2 + x - 6 = x(x + 1) - 6$
(D) $x^2 + x - 6 = (x - 1)(x + 6)$
(E) $x^2 + x - 6 = x^2 + x$
- Q6.** Factor the trinomial $x^2 + 10x + 24$.
(A) $x^2 + 10x + 24 = (x + 4)(x + 6)$
(B) $x^2 + 10x + 24 = (x + 2)(x + 12)$
(C) $x^2 + 10x + 24 = x(x + 10) + 24$
(D) $x^2 + 10x + 24 = (x + 1)(x + 24)$
(E) $x^2 + 10x + 24 = (x + 3)(x + 8)$
- Q7.** Factor the trinomial $x^2 - 9x + 20$.
(A) $x^2 - 9x + 20 = (x - 4)(x - 5)$
(B) $x^2 - 9x + 20 = (x - 1)(x - 20)$
(C) $x^2 - 9x + 20 = (x - 2)(x - 10)$
(D) $x^2 - 9x + 20 = x(x - 9) + 20$
(E) $x^2 - 9x + 20 = (x - 1)(x - 8)$
- Q8.** A rectangular garden measures $x^2 + 5x + 6$ feet. Factor the expression to find the length and width of the garden.
(A) $x^2 + 5x + 6 = (x + 2)(x + 3)$
(B) $x^2 + 5x + 6 = (x + 1)(x + 6)$
(C) $x^2 + 5x + 6 = x(x + 5) + 6$
(D) $x^2 + 5x + 6 = (x + 1)(x + 4)$
(E) $x^2 + 5x + 6 = x^2 + 5x$

Open Ended Questions (FRQ)

- Q9.** Factor the trinomial $2x^2 + 5x - 3$. Show all steps and simplify your answer.
- Q10.** A water tank can hold $x^2 - 4x - 5$ gallons of water. Factor the expression to determine the dimensions of the tank. Show all steps and simplify your answer.

Chef's Tips

- Tip 1: Make sure to check the units of the answer.
- Tip 2: Consider using different factoring methods for each problem.
- Tip 3: Use the context of the problem to help with factoring.